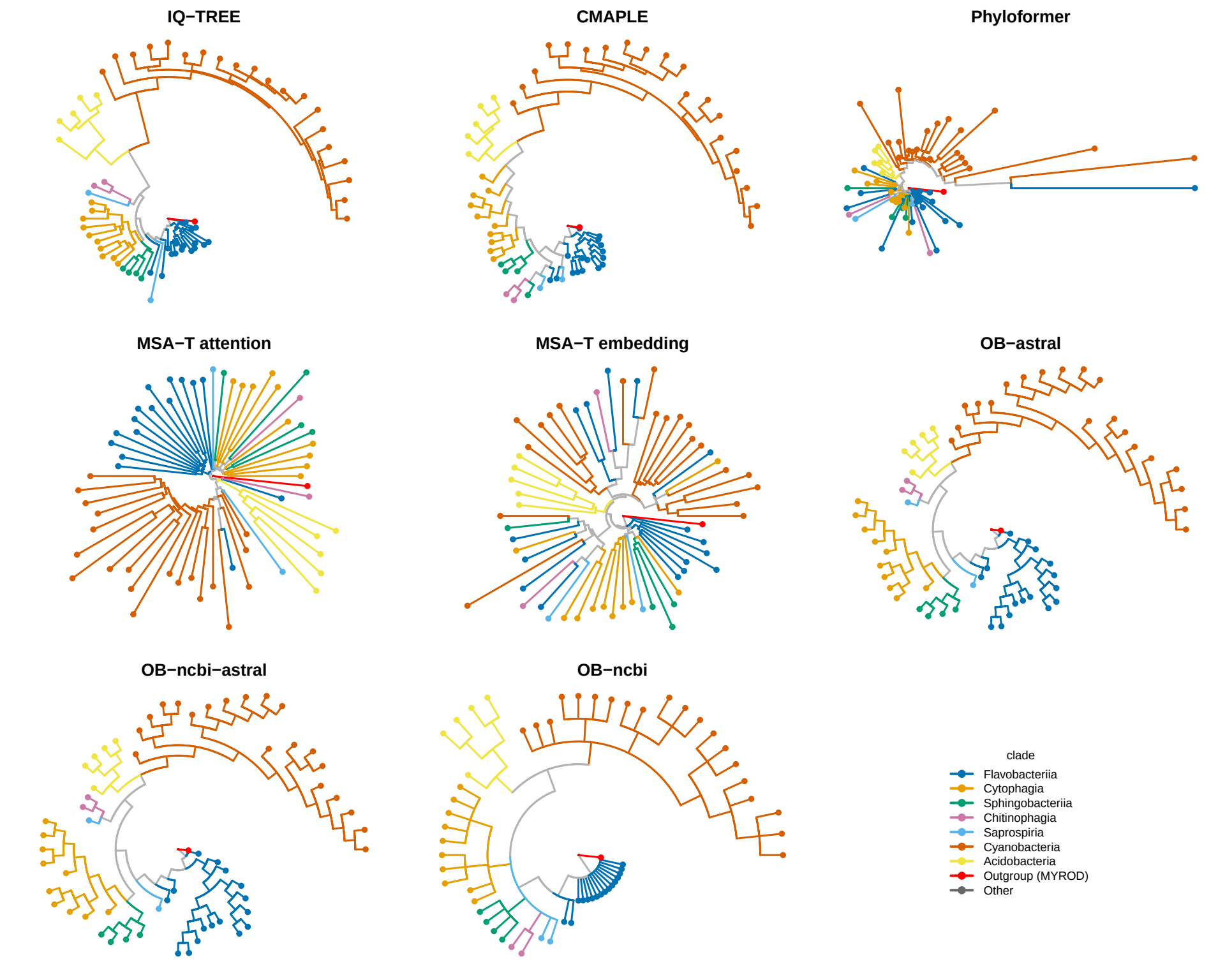
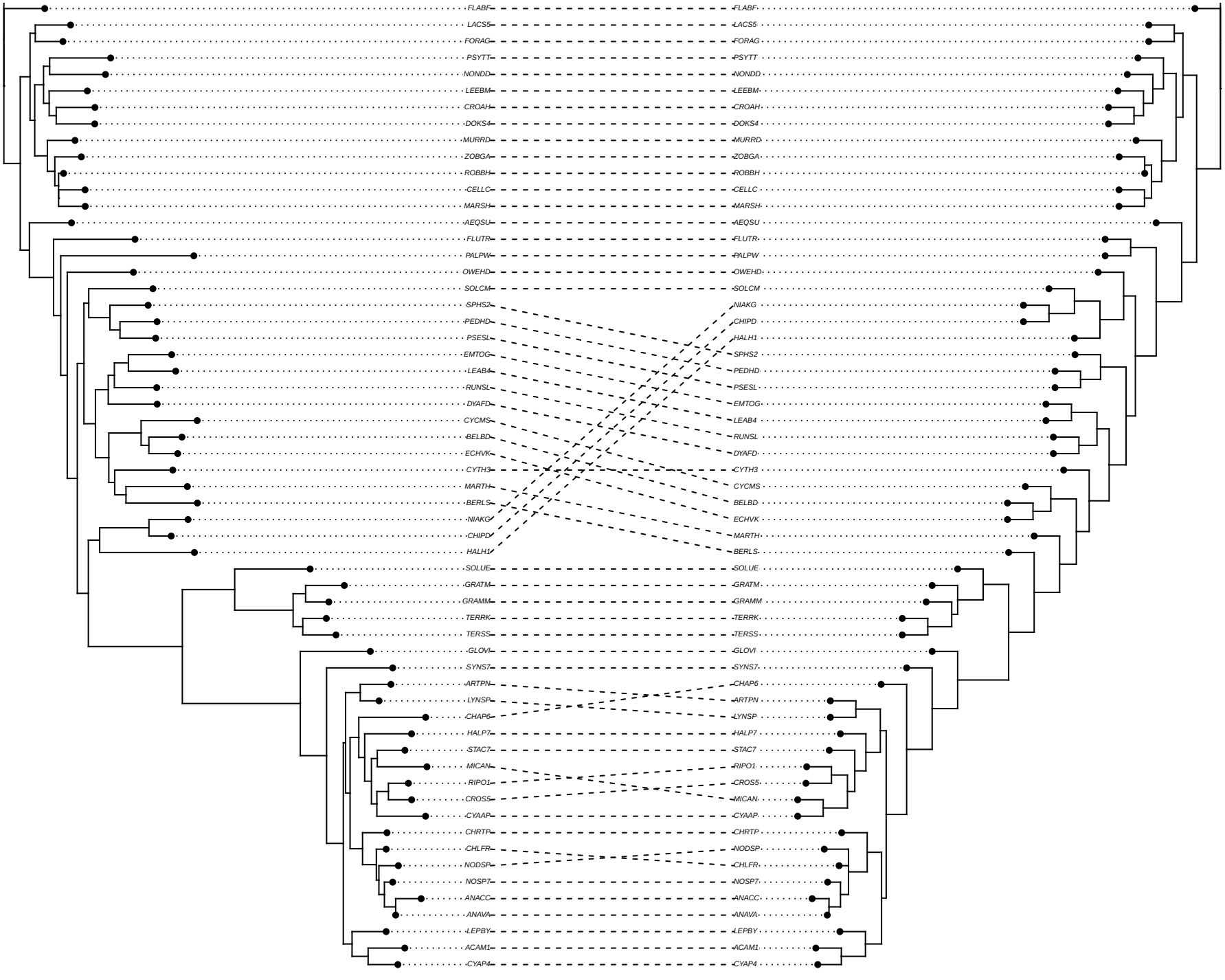
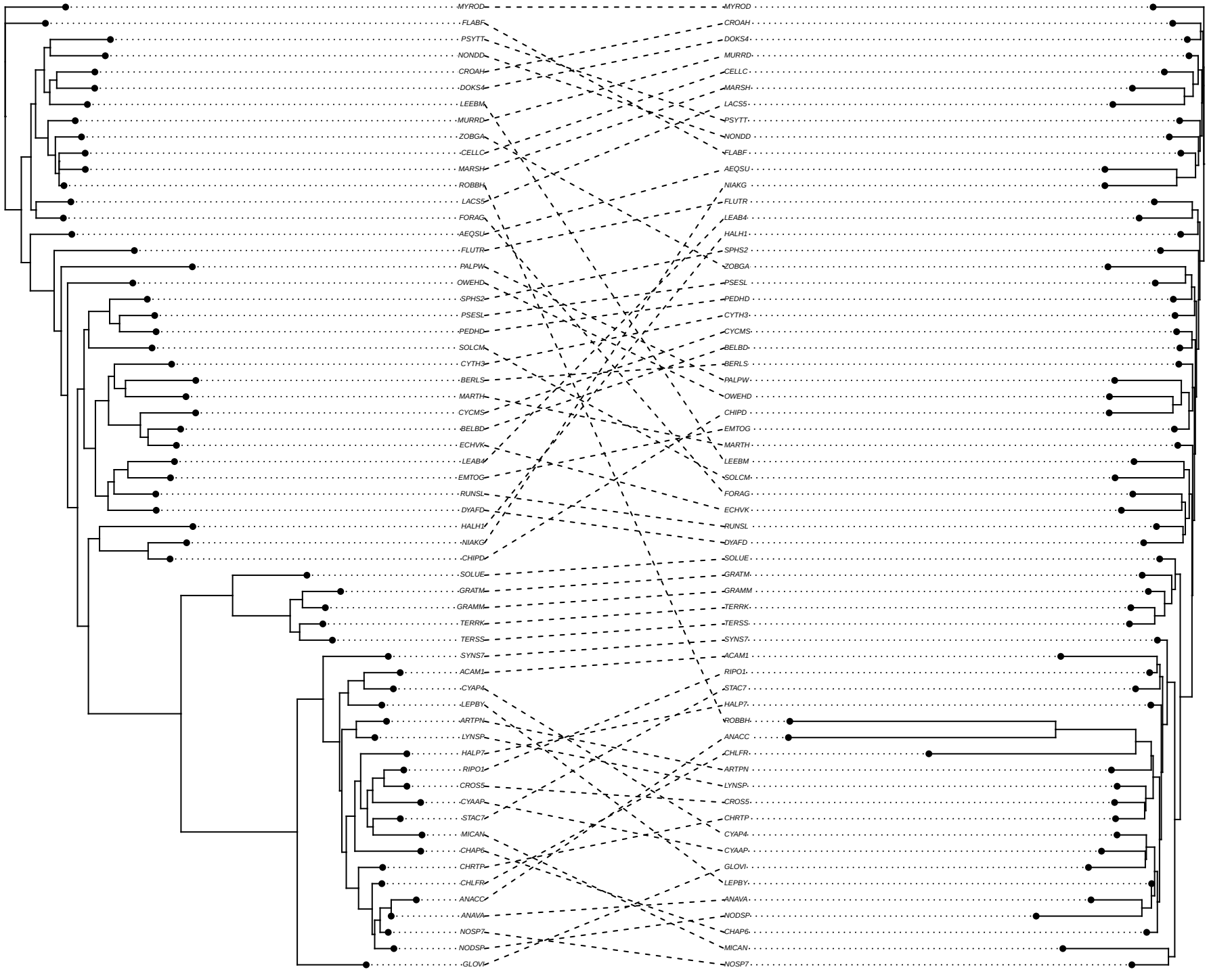
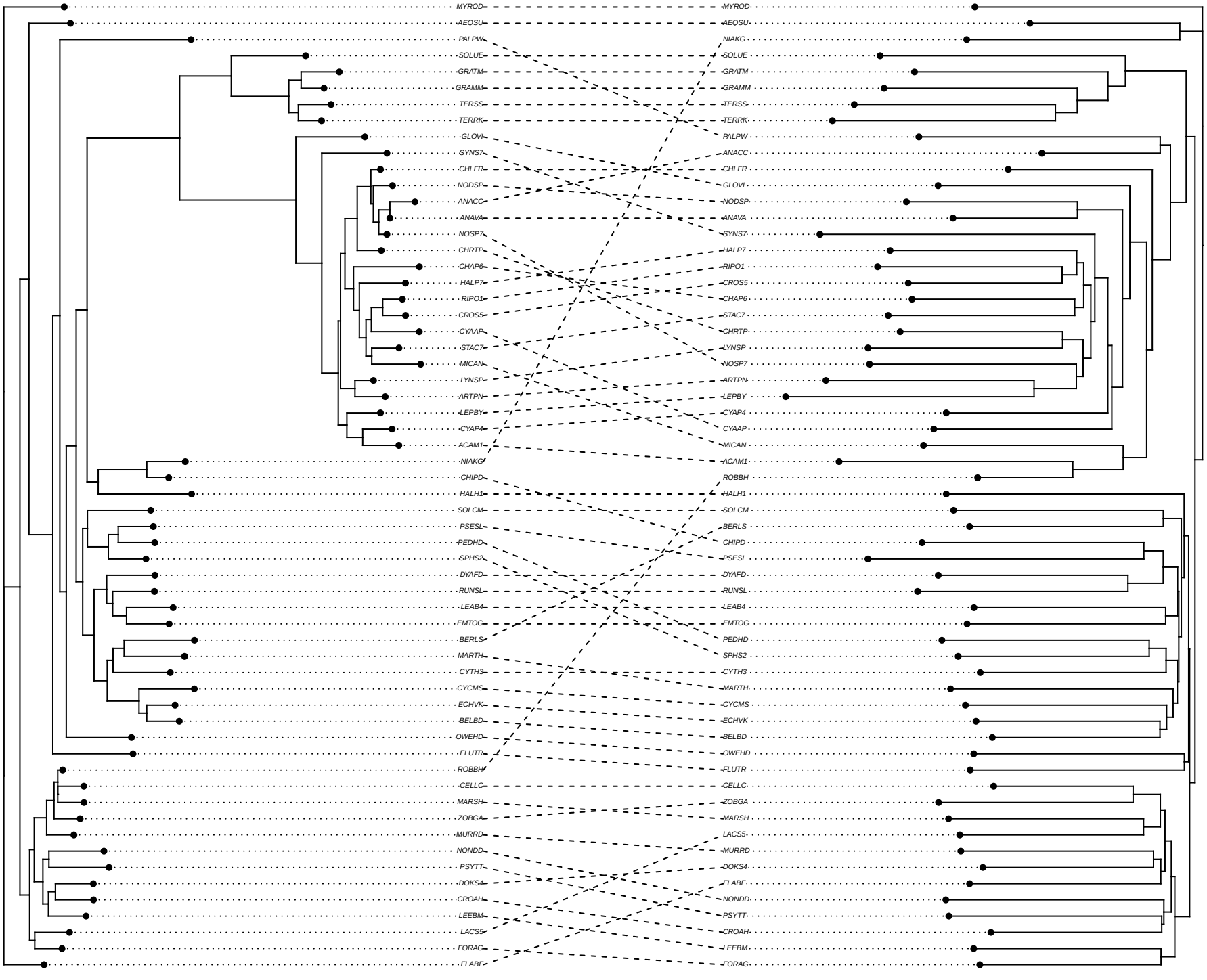
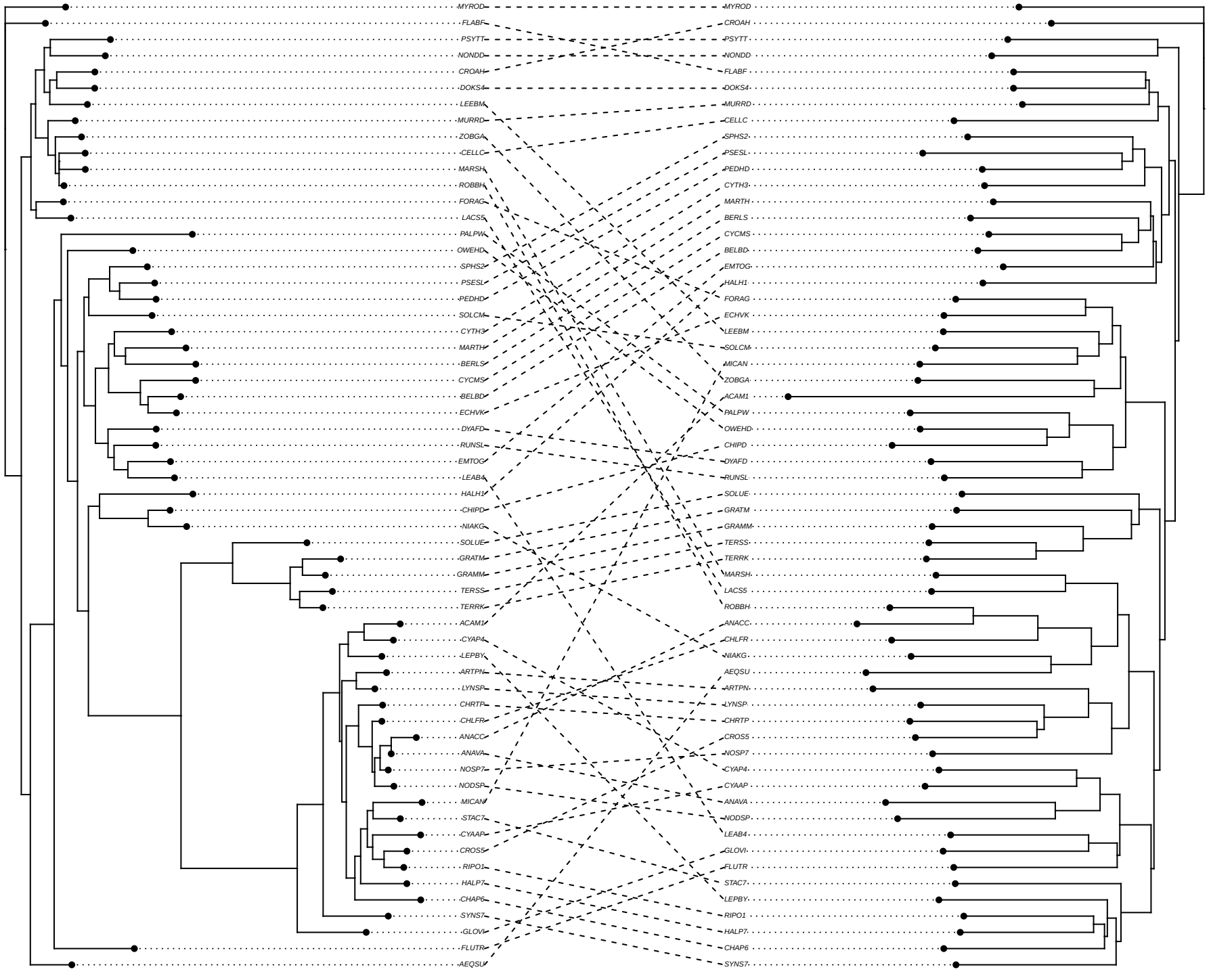

IQ−TREE
CMAPLE
Phyloformer
MSA−T attention
MSA−T embedding
OB−astral
OB−ncbi−astral
OB−ncbi
clade
Flavobacteriia
Cytophagia
Sphingobacteriia
Chitinophagia
Saprospiria
Cyanobacteria
Acidobacteria
Outgroup (MYROD)
Other

IQ-TREE vs CMAPLE

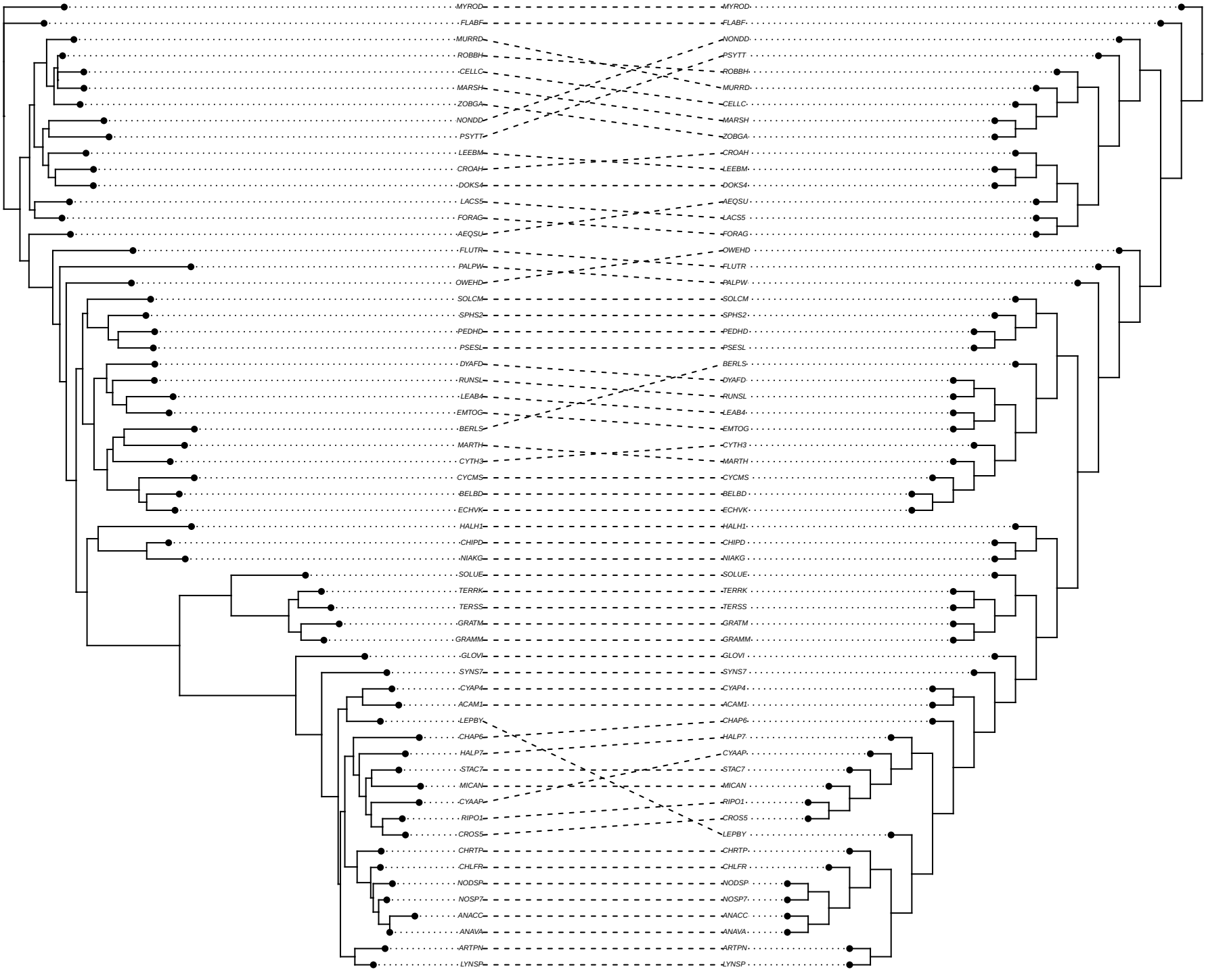



IQ−TREE vs Phylolorimer
MYROD
FLABF
PSYTT
NONDD
CROAH
DOKS4
LEEBM
MURRD
ZOBGA
CELLC
MARSH
ROBBH
LACS5
FORAG
AEQSU
FLUTR
PALPW
OWEHD
SPHS2
PSESL
PEDHD
SOLCM
CYTH3
BERLS
MARTH
CYCMS
BELBD
ECHVK
LEAB4
EMTOG
RUNSL
DYAFD
HALH1
NIAKG
CHIPD
SOLUE
GRATM
GRAMM
TERRK
TERSS
SYNS7
ACAM1
CYAP4
LEPBY
ARTPN
LYNSP
HALP7
RIPO1
CROS5
CYAAP
STAC7
MICAN
CHAP6
CHRTP
CHLFR
ANACC
ANAVA
NOSP7
NODSP
GLOVI
MYROD
CROAH
DOKS4
MURRD
CELLC
MARSH
LACS5
PSYTT
NONDD
FLABF
AEQSU
NIAKG
FLUTR
LEAB4
HALH1
SPHS2
ZOBGA
PSESL
PEDHD
CYTH3
CYCMS
BELBD
BERLS
PALPW
OWEHD
CHIPD
EMTOG
MARTH
LEEBM
SOLCM
FORAG
ECHVK
RUNSL
DYAFD
SOLUE
GRATM
GRAMM
TERRK
TERSS
SYNS7
ACAM1
RIPO1
STAC7
HALP7
ROBBH
ANACC
CHLFR
ARTPN
LYNSP
CROS5
CHRTP
CYAP4
CYAAP
GLOVI
LEPBY
ANAVA
NODSP
CHAP6
MICAN
NOSP7

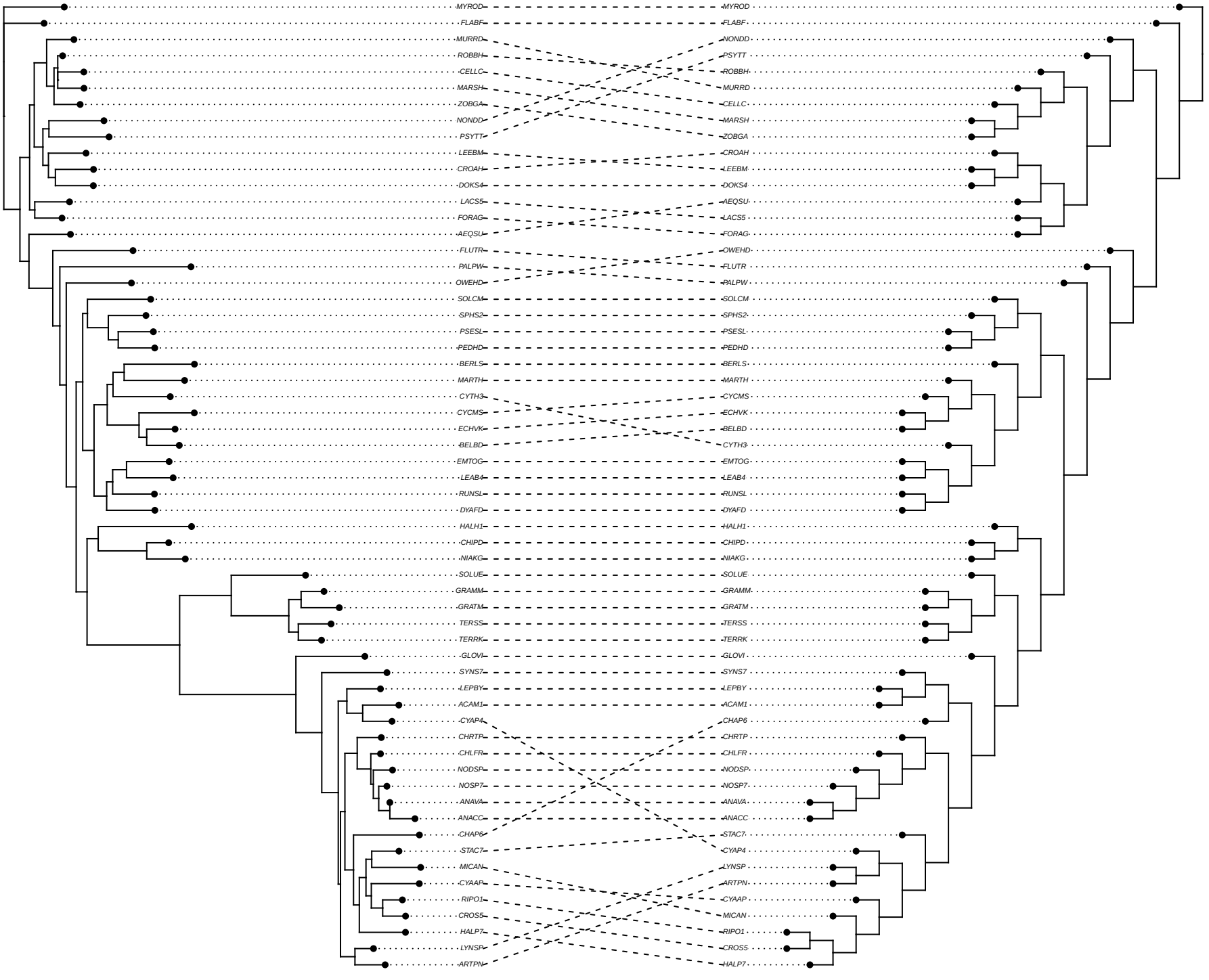

IQ−TREE vs MSA−Pattention
MYROD
AEQSU
PALPW
SOLUE
GRATM
GRAMM
TERSS
TERRK
GLOVI
SYNS7
CHLFR
NODSP
ANACC
ANAVA
NOSP7
CHRTP
CHAP6
HALP7
RIPO1
CROS5
CYAAP
STAC7
MICAN
LYNSP
ARTPN
LEPBY
CYAP4
ACAM1
NIAKG
CHIPD
HALH1
SOLCM
PSESL
PEDHD
SPHS2
DYAFD
RUNSL
LEAB4
EMTOG
BERLS
MARTH
CYTH3
CYCMS
ECHVK
BELBD
OWEHD
FLUTR
ROBBH
CELLC
MARSH
ZOBGA
MURRD
NONDD
PSYTT
DOKS4
CROAH
LEEBM
LACS5
FORAG
FLABF
MYROD
AEQSU
NIAKG
SOLUE
GRATM
GRAMM
TERSS
TERRK
PALPW
ANACC
CHLFR
GLOVI
NODSP
ANAVA
SYNS7
HALP7
RIPO1
CROS5
CHAP6
STAC7
CHRTP
LYNSP
NOSP7
ARTPN
LEPBY
CYAP4
CYAAP
MICAN
ACAM1
ROBBH
HALH1
SOLCM
BERLS
CHIPD
PSESL
DYAFD
RUNSL
LEAB4
EMTOG
PEDHD
SPHS2
CYTH3
MARTH
CYCMS
ECHVK
BELBD
OWEHD
FLUTR
CELLC
ZOBGA
MARSH
LACS5
MURRD
DOKS4
FLABF
NONDD
PSYTT
CROAH
LEEBM
FORAG

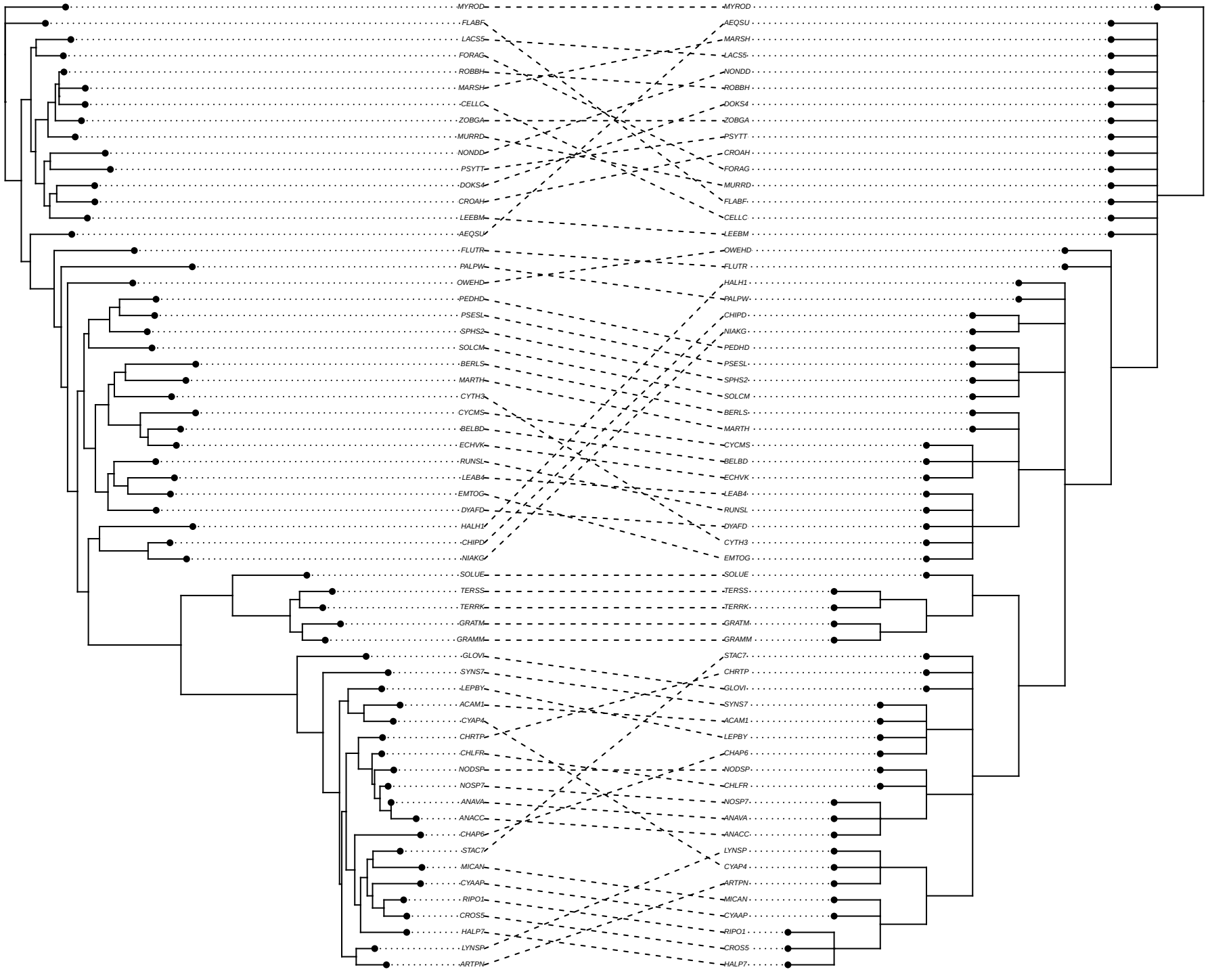

IQ−TREE vs MSA−T embedding
MYROD
FLABF
PSYTT
NONDD
CROAH
DOKS4
LEEBM
MURRD
ZOBGA
CELLC
MARSH
ROBBH
FORAG
LACS5
PALPW
OWEHD
SPHS2
PSESL
PEDHD
SOLCM
CYTH3
MARTH
BERLS
CYCMS
BELBD
ECHVK
DYAFD
RUNSL
EMTOG
LEAB4
HALH1
CHIPD
NIAKG
SOLUE
GRATM
GRAMM
TERSS
TERRK
ACAM1
CYAP4
LEPBY
ARTPN
LYNSP
CHRTP
CHLFR
ANACC
ANAVA
NOSP7
NODSP
MICAN
STAC7
CYAAP
CROS5
RIPO1
HALP7
CHAP6
SYNS7
GLOVI
FLUTR
AEQSU
MYROD
CROAH
PSYTT
NONDD
FLABF
DOKS4
MURRD
CELLC
SPHS2
PSESL
PEDHD
CYTH3
MARTH
BERLS
CYCMS
BELBD
EMTOG
HALH1
FORAG
ECHVK
LEEBM
SOLCM
MICAN
ZOBGA
ACAM1
PALPW
OWEHD
CHIPD
DYAFD
RUNSL
SOLUE
GRATM
GRAMM
TERSS
TERRK
MARSH
LACS5
ROBBH
ANACC
CHLFR
NIAKG
AEQSU
ARTPN
LYNSP
CHRTP
CROS5
NOSP7
CYAP4
CYAAP
ANAVA
NODSP
LEAB4
GLOVI
FLUTR
STAC7
LEPBY
RIPO1
HALP7
CHAP6
SYNS7

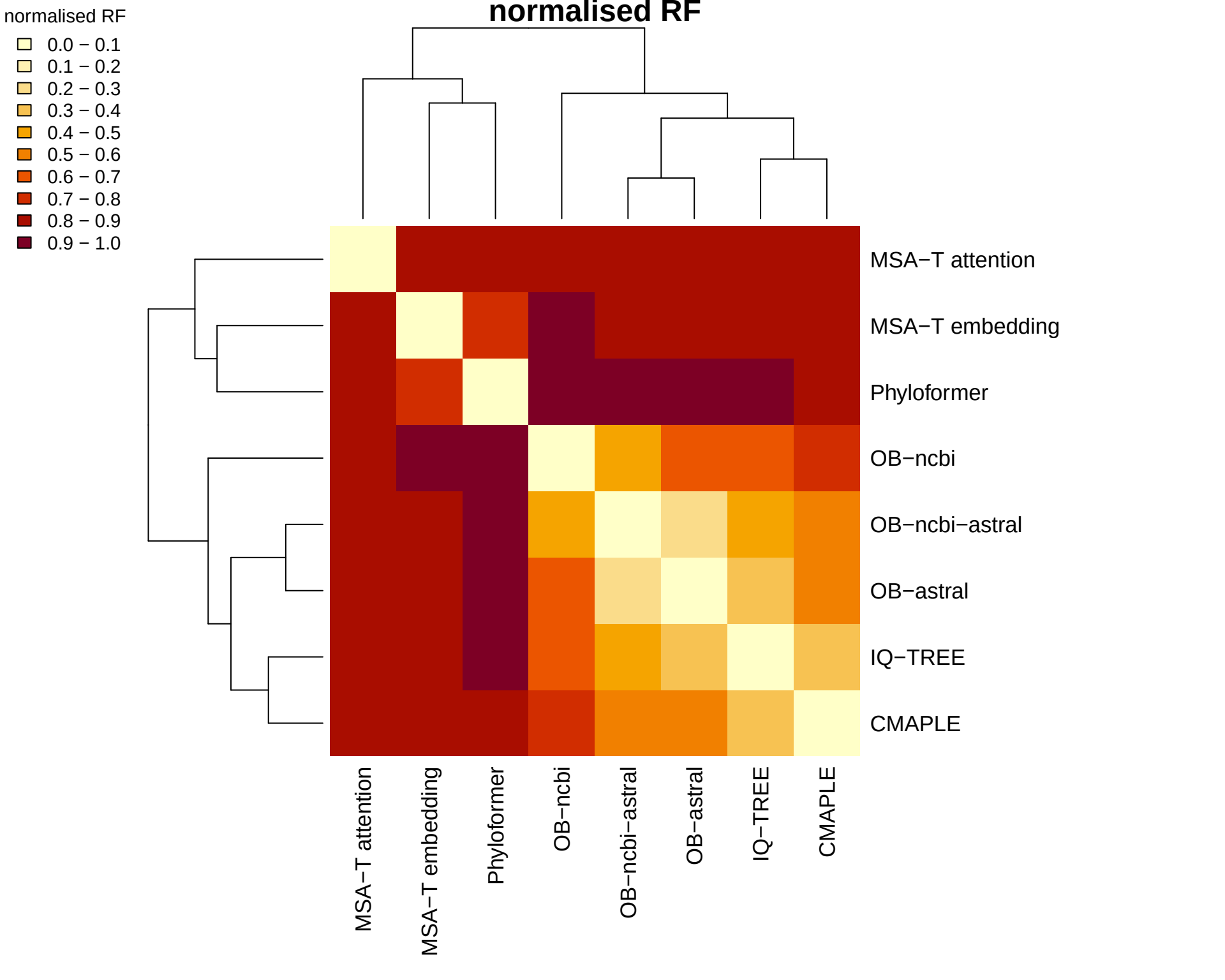
IQ−TREE vs OB−astral

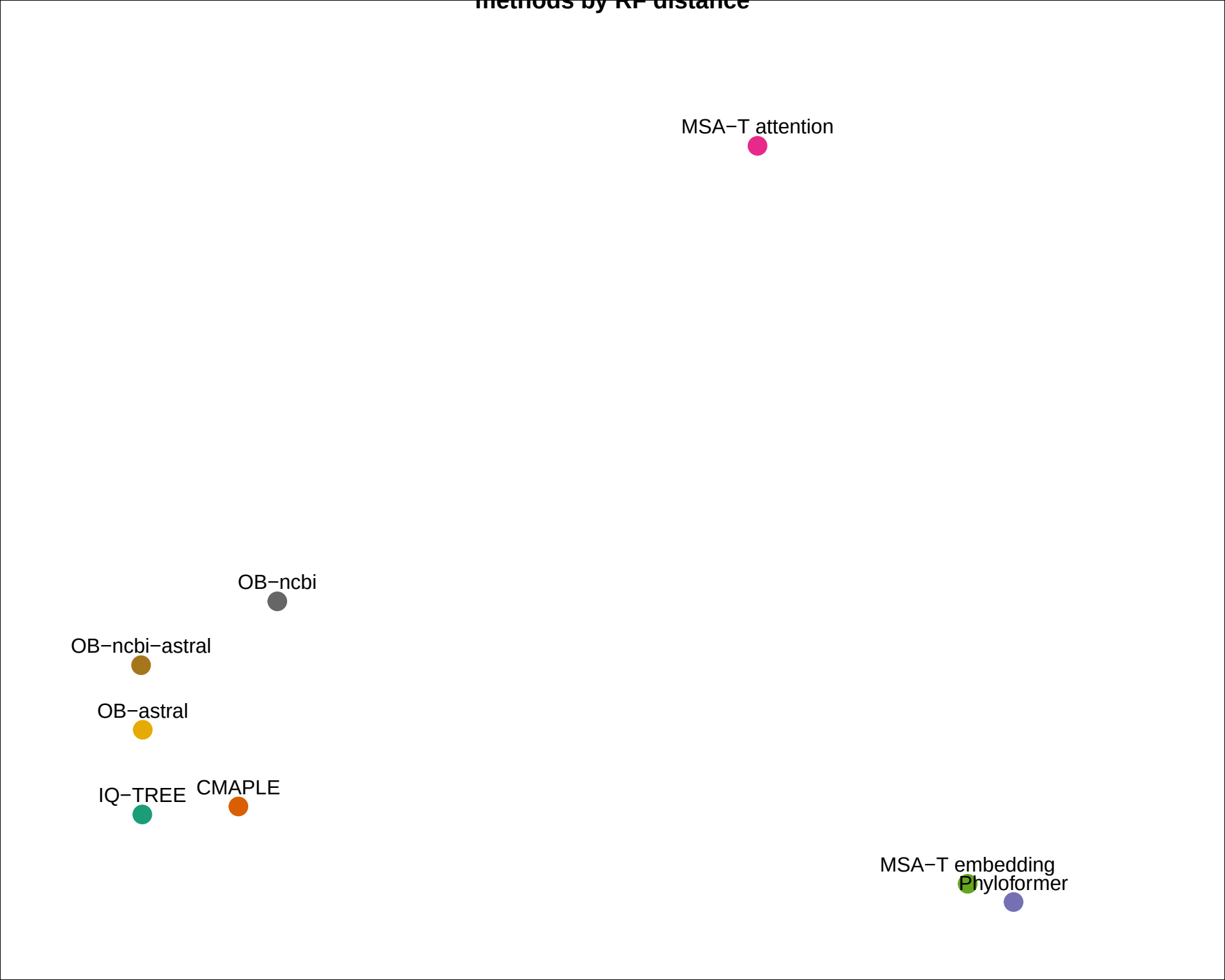


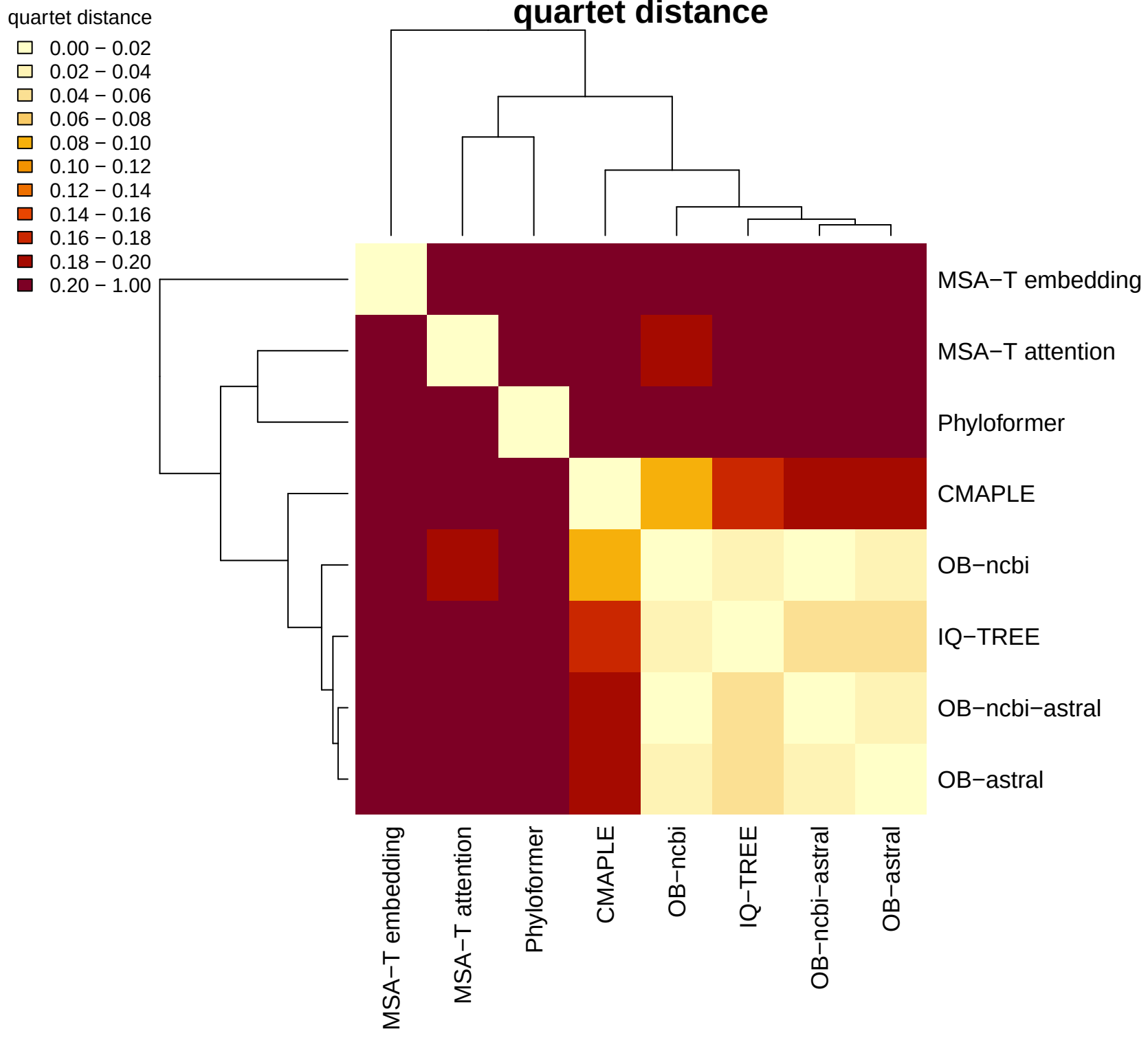
IQ-TREE vs OB-ncbi-astral

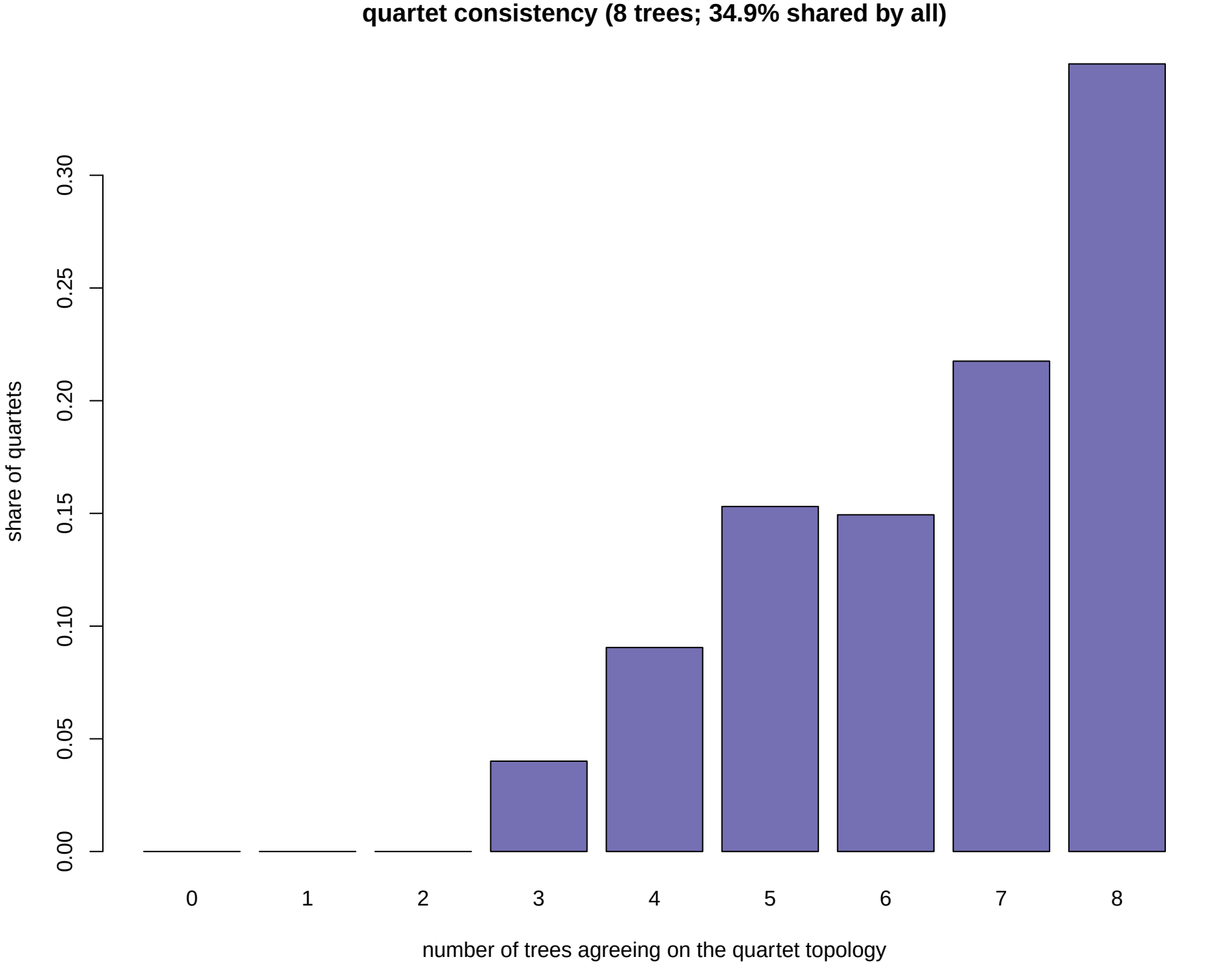


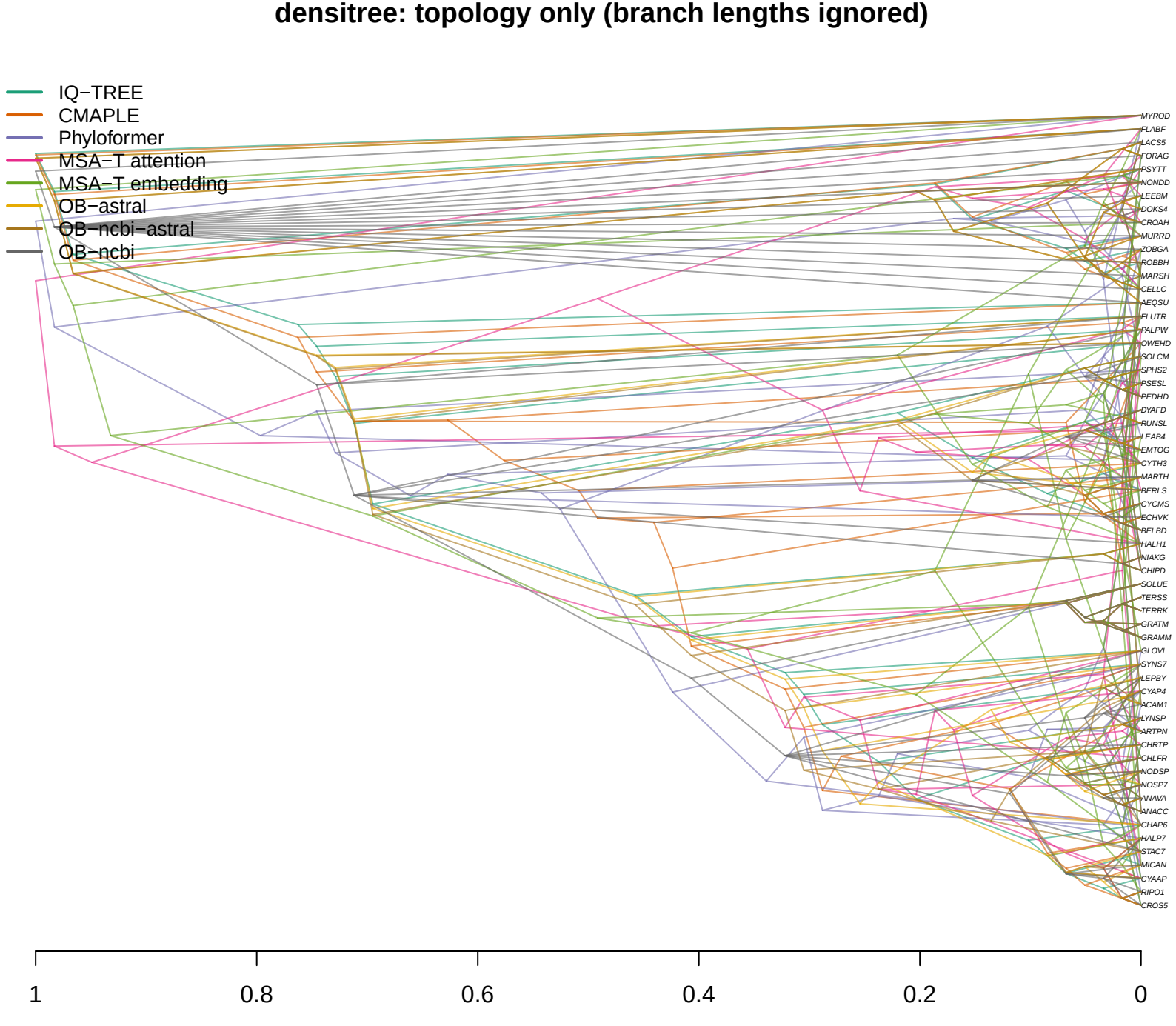

IQ−TREE vs OB−ncbi
MYROD
FLABF
LACS5
FORAG
ROBBH
MARSH
CELLC
ZOBGA
MURRD
NONDD
PSYTT
DOKS4
CROAH
LEEBM
AEQSU
FLUTR
PALPW
OWEHD
PEDHD
PSESL
SPHS2
SOLCM
BERLS
MARTH
CYTH3
CYCMS
BELBD
ECHVK
RUNSL
LEAB4
EMTOG
DYAFD
HALH1
CHIPD
NIAKG
SOLUE
TERSS
TERRK
GRATM
GRAMM
GLOVI
SYNS7
LEPBY
ACAM1
CYAP4
CHRTP
CHLFR
NODSP
NOSP7
ANAVA
ANACC
CHAP6
STAC7
MICAN
CYAAP
RIPO1
CROS5
HALP7
LYNSP
ARTPN
MYROD
AEQSU
MARSH
LACS5
NONDD
ROBBH
DOKS4
ZOBGA
PSYTT
CROAH
FORAG
MURRD
FLABF
CELLC
LEEBM
OWEHD
FLUTR
HALH1
PALPW
CHIPD
NIAKG
PEDHD
PSESL
SPHS2
SOLCM
BERLS
MARTH
CYCMS
BELBD
ECHVK
LEAB4
RUNSL
DYAFD
CYTH3
EMTOG
SOLUE
TERSS
TERRK
GRATM
GRAMM
STAC7
CHRTP
GLOVI
SYNS7
ACAM1
LEPBY
CHAP6
NODSP
CHLFR
NOSP7
ANAVA
ANACC
LYNSP
CYAP4
ARTPN
MICAN
CYAAP
RIPO1
CROS5
HALP7

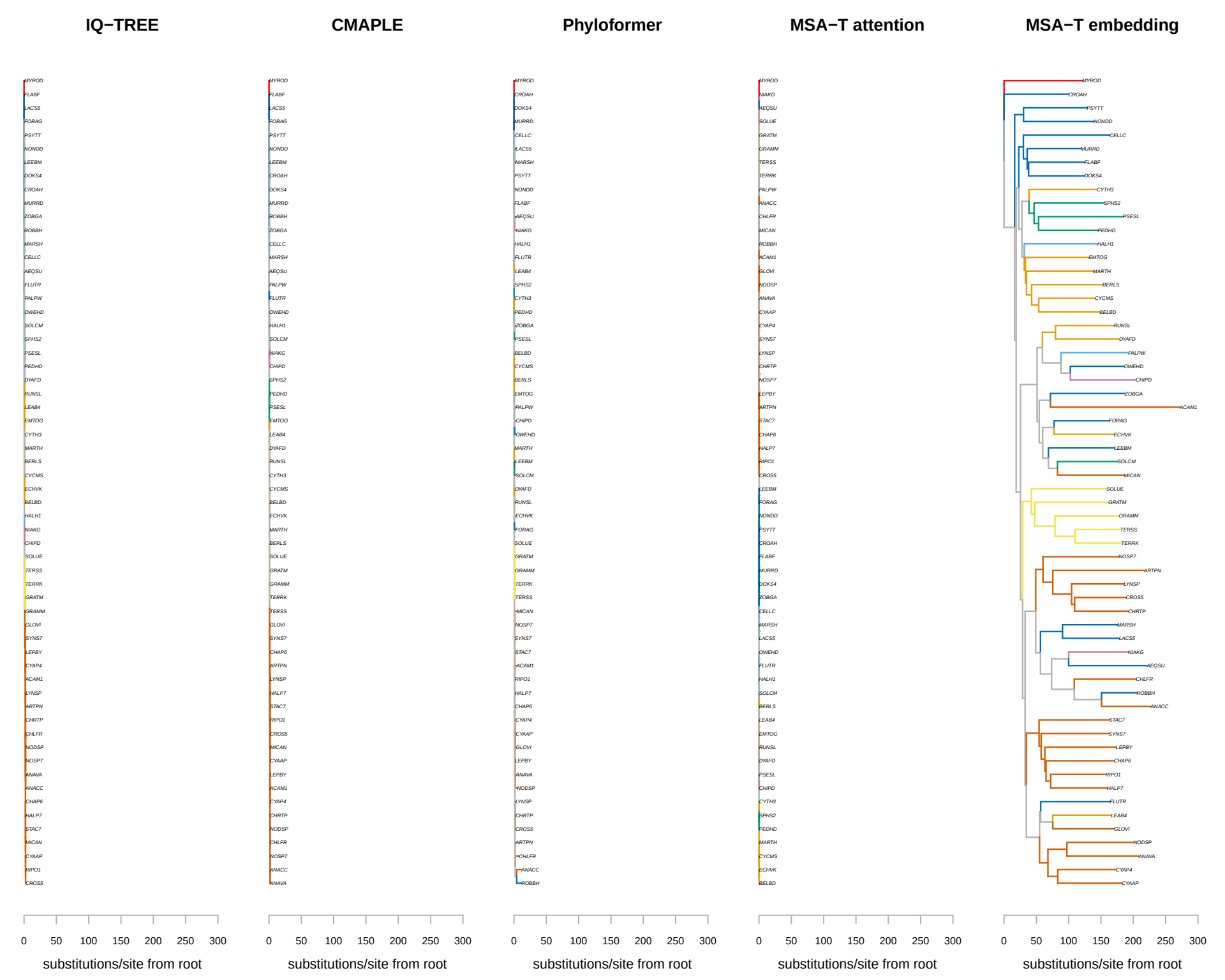

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T attention
MSA−T embedding
Phyloformer
OB−ncbi
OB−ncbi−astral
OB−astral
IQ−TREE
CMAPLE
MSA−T attention
MSA−T embedding
Phyloformer
OB−ncbi
OB−ncbi−astral
OB−astral
IQ−TREE
CMAPLE


methods by RF distance
MSA−T attention
OB−ncbi
OB−ncbi−astral
OB−astral
IQ−TREE
CMAPLE
MSA−T embedding
Phyloformer

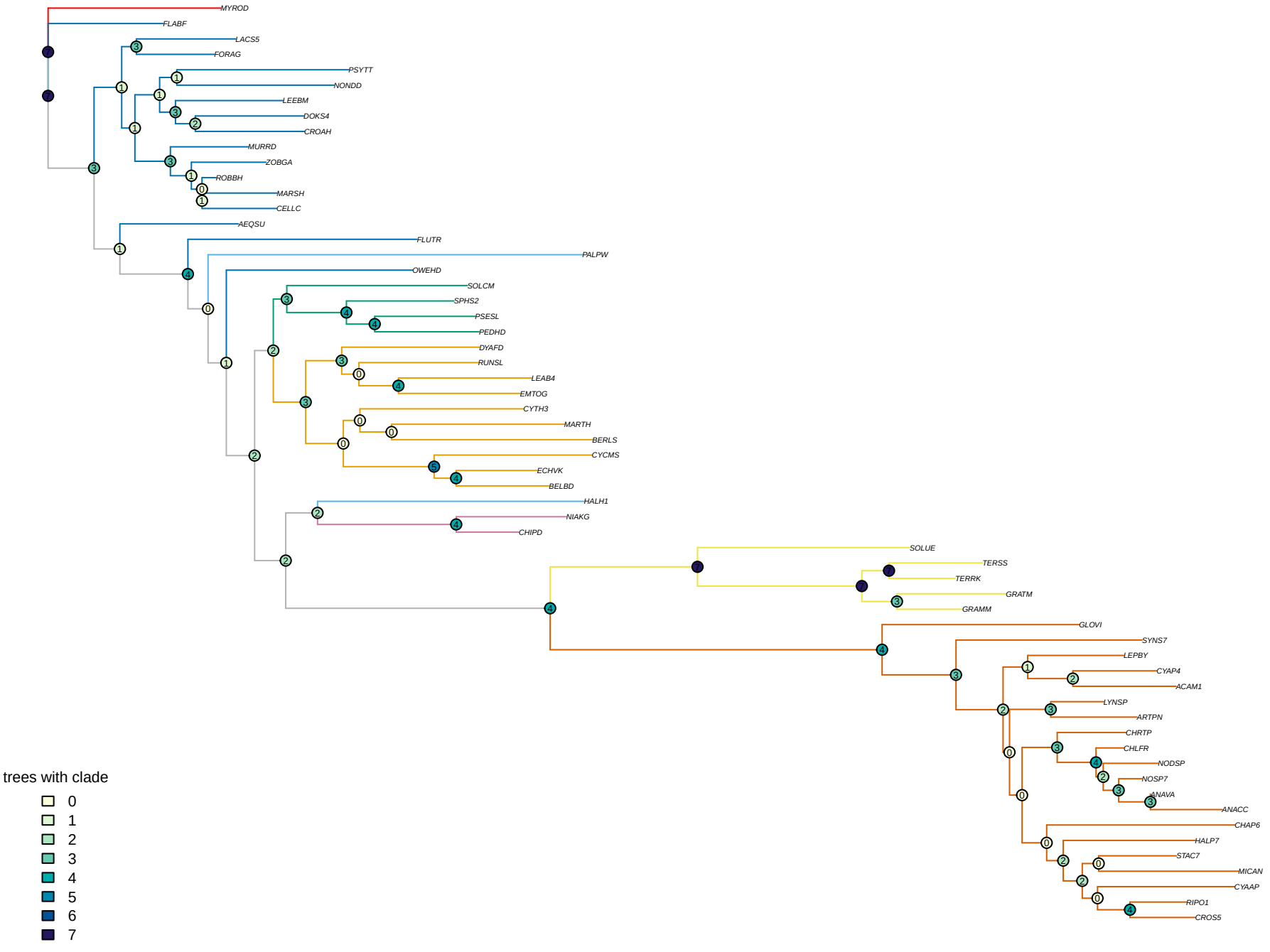

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
MSA−T embedding
MSA−T attention
Phyloformer
CMAPLE
OB−ncbi
IQ−TREE
OB−ncbi−astral
OB−astral
MSA−T embedding
MSA−T attention
Phyloformer
CMAPLE
OB−ncbi
IQ−TREE
OB−ncbi−astral
OB−astral


quartet consistency (8 trees; 34.9% shared by all)
share of quartets
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

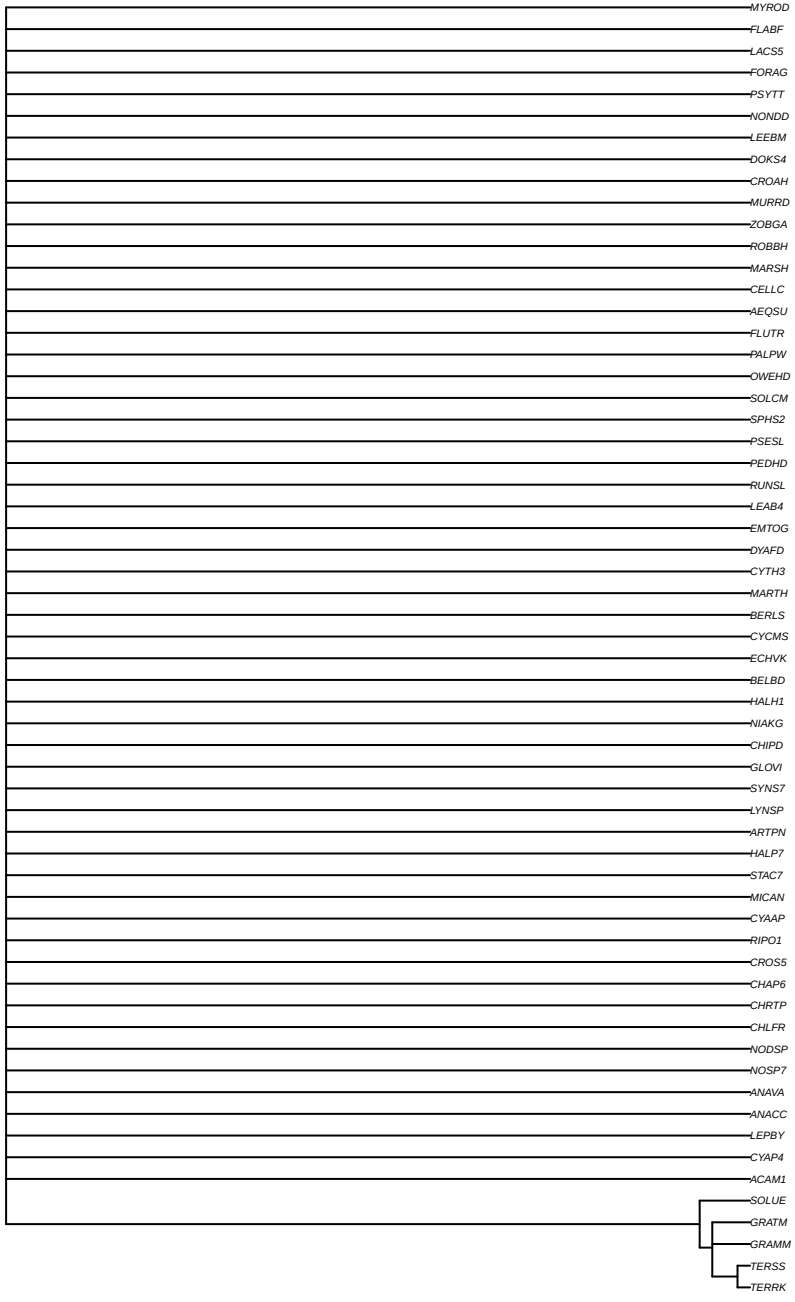

densitree: topology only (branch lengths ignored)
IQ−TREE
CMAPLE
Phyloformer
MSA−T attention
MSA−T embedding
OB−astral
OB−ncbi−astral
OB−ncbi
MYROD
FLABF
LACS5
FORAG
PSYTT
NONDD
LEEBM
DOKS4
CROAH
MURRD
ZOBGA
ROBBH
MARSH
CELLC
AEQSU
FLUTR
PALPW
OWEHD
SOLCM
SPHS2
PSESL
PEDHD
DYAFD
RUNSL
LEAB4
EMTOG
CYTH3
MARTH
BERLS
CYCMS
ECHVK
BELBD
HALH1
NIAKG
CHIPD
SOLUE
TERSS
TERRK
GRATM
GRAMM
GLOVI
SYNS7
LEPBY
CYAP4
ACAM1
LYNSP
ARTPN
CHRTP
CHLFR
NODSP
NOSP7
ANAVA
ANACC
CHAP6
HALP7
STAC7
MICAN
CYAAP
RIPO1
CROS5
1
0.8
0.6
0.4
0.2
0


IQ−TREE
CMAPLE
Phyloformer
MSA−T attention
MSA−T embedding
MYROD
FLABF
LACS5
FORAG
PSYTT
NONDD
LEEBM
DOKS4
CROAH
MURRD
ZOBGA
ROBBH
MARSH
CELLC
AEQSU
FLUTR
PALPW
OWEHD
SOLCM
SPHS2
PSESL
PEDHD
DYAFD
RUNSL
LEAB4
EMTOG
CYTH3
MARTH
BERLS
CYCMS
ECHVK
BELBD
HALH1
NIAKG
CHIPD
SOLUE
TERSS
TERRK
GRATM
GRAMM
GLOVI
SYNS7
LEPBY
CYAP4
ACAM1
LYNSP
ARTPN
CHRTP
CHLFR
NODSP
NOSP7
ANAVA
ANACC
CHAP6
HALP7
STAC7
MICAN
CYAAP
RIPO1
CROSS5
0
50
100
150
200
250
300
substitutions/site from root

IQ−TREE clades: number of the other 7 trees that contain each clade



strict



majority rule

